

Herald



Herald — AGENTS.md

Field	Value
Revision	17
Created	2026-05-15
Last modified	2026-05-31
Status	active
Status summary	r17: refreshed the stale "pre-implementation (2026-05-15) / Go-scaffold landed, 5 modules" project-status prose to current reality — 18 workspace modules (10 shared/foundation + 8 flavor binaries), waves 2–7 shipped, tag v0.6.0 (2026-05-28), live pherald listen inbound runtime + Claude Code dispatch + Telegram & Slack adapters + REST API + participant-attribution (§11.4.104) + intent-recognition (§11.4.105) + Docs Chain (§11.4.106); HRD-010/011/012/016 now LIVE; ~a-dozen open HRDs remain. Status prose ONLY — covenant/build/anti-bluff/inheritance content unchanged (merged from a parallel stream's worktree; took main's r16 §11.4.106 header + the stream's project-status prose). Prior r16: propagated HelixConstitution §11.4.106 (Docs Chain documentation-sync mandate, 2026-05-29) into the agent-rule inherited-covenant cluster as a short-form **§11.4.106 ...** restatement citing the literal anchor 11.4.106, inherited per §11.4.35; also added the cluster-header's previously-omitted §11.4.100. Herald-applicability APPLICABLE (in progress) — cites docs/research/docs_chain/HERALD_DOCS_CHAIN_PLAN.md, the planned E146 drift-gate, and the open prerequisites (G6 exec-staging, dogfood binary-hash verify fix in flight); restated + cited, not redefined. Prior r15: synced the four spec pointers in this file (ToC + project-status bullet + spec-change-rule heading + rule body) from the now-archived docs/specs/mvp/specification.V3.md to the active docs/specs/mvp/specification.V4.md. V4 superseded V3 on 2026-05-31 (V3 archived under docs/specs/mvp/archive/); the V4-creation stream updated CLAUDE.md/README/Status.md but missed AGENTS.md, leaving broken pointers to a moved file. Pure path-sync — no rule semantics changed; inheritance gate 15/15 PASS (I7b spec-change anchor intact). Companion fixes same effort: cherald spec-version-check default V3→V4 (checkops_cmds.go, verified on the real binary) + its guide, Status_Summary.md active_spec drift (V3→V4, +V3 archived), HRD-092 catalogue-check pointed at the archived V3. Prior r14: propagated the intent-recognition mandate (operator 2026-05-31) into the agent-rule cluster as a short-form restatement — new "Intent recognition & clarification" section + ToC entry — citing the authoritative contract docs/design/

Field	Value
	INTENT_RECOGNITION.md and noting inheritance from HelixConstitution §11.4.105 (the root-§ being added on the constitution stream), inherited per §11.4.35. Covers: subscribers speak plain natural language (NO command syntax / no COMMAND: prefix); three-tier resolution (Tier 1 deterministic CommandRecognizer fast-path → Tier 2 LLM intent inference via the <<<HERALD-DISPATCH-v1>>> envelope → Tier 3 action="clarify" reply-tag-and-ask fallback via the §11.4.104 IdentityResolver); never guess an action, never ignore a message. Restated + cited, not redefined. Prior r13: propagated the participant/attribution/tagging mandate (operator 2026-05-31) into the agent-rule cluster as a short-form restatement — new "Participant identity, attribution & notification-tagging" section + ToC entry — citing the authoritative contract docs/design/PARTICIPANT_ATTRIBUTION.md and noting inheritance from HelixConstitution per §11.4.35. Covers per-messenger Participant identity (subscribers/subscriber_aliases.username), the HERALD_<CHANNEL>_OPERATOR_USERNAME operator env var (HERALD_TGRAM_OPERATOR_USERNAME=@milos85vasic), created_by/assigned_to attribution, and the @-tagging matrix (Claude + Operator never tagged). Restated + cited, not redefined. Prior r12: propagated HelixConstitution §11.4.100 (Video color + visual-quality fidelity mandate, 2026-05-28) into the agent-rule covenant cluster as a short-form restatement citing the literal anchor 11.4.100, inherited per §11.4.35; restated + cited, not redefined. Herald-applicability: non-applicable-but-cite (Herald has NO video-playback surface — pherald listen downloads video attachments as opaque sha256-blobs without decoding/rendering); mandate binds latently if Herald ever ships a video-rendering surface. Cascade-parallel: §11.4.96 ("Herald has no AOSP build, but the principle binds"). Required by upcoming CM-COVENANT-114-100-PROPAGATION pre-build gate. Prior r9: propagated HelixConstitution §11.4.95–§11.4.97 (workable-items SQLite DB tracked-in-git-not-gitignored, safe-parallel-work-with-long-build catalogue + mandate, maximum-use-of-idle-time + progress-update cadence) into the agent-rule covenant cluster as short-form restatements citing the literal anchors 11.4.95/11.4.96/11.4.97, inherited per §11.4.35; restated + cited, not redefined. Prior r8: propagated HelixConstitution §11.4.89–§11.4.94 (background-test execution, Obsolete status + obsolescence audit, summary-doc clarity, multi-pass change-evaluation, SQLite workable-items SSoT, zero-idle parallel-by-default) into the agent-rule covenant cluster as short-form restatements citing the literal anchors, inherited per §11.4.35; restated + cited, not redefined. Prior r7: propagated HelixConstitution §11.4.85 (stress + chaos test mandate) + §11.4.87 (endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing) + §11.4.88 (background-push) into the agent-rule covenant cluster as short-form restatements citing the literal section numbers, inherited per §11.4.35; required by the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate (literal anchor 11.4.87 MUST be present in every CLAUDE.md/AGENTS.md/QWEN.md; same expectation for §11.4.85/§11.4.88 per cascade pattern). Agents restate + cite, never redefine or weaken. Prior r5: added the End-user-usability covenant section restating the verbatim operator mandate at Herald agent-rule level; binds every CLI agent (Claude Code, Codex, Cursor, Gemini, Aider, subagents) to the §11.4 anti-bluff enforcement; ties to HERALD_CONSTITUTION.md §107 + inheritance-gate invariant I8b.
Issues	none
Issues summary	—

Field	Value
Fixed	R-14 (V2), V3-path-sync (V3 r3), Go-scaffold-status-update (V3 r4), §107 mandate restatement + I8b anchor (r5), Helix §11.4.85 + §11.4.87 + §11.4.88 propagation (r7), Helix §11.4.89–§11.4.94 propagation (r8), Helix §11.4.95–§11.4.97 propagation (r9), spec-V4-path-sync (r15), Helix §11.4.106 Docs-Chain propagation (r16)
Fixed summary	aligned with HRD-009/HRD-009b/HRD-013/HRD-014 landing in the same commit; r5 closes the Herald-level explicit-restatement gap identified by the 2026-05-20 audit; r7 propagates the three new inherited HelixConstitution mandates (§11.4.85 stress+chaos, §11.4.87 endless-loop autonomous work + zero-idle dispatch + anti-bluff testing, §11.4.88 background-push) into this agent-rules file per the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate, inherited per §11.4.35; restated + cited, not redefined; r8 propagates the next six inherited HelixConstitution mandates (§11.4.89 background-test execution, §11.4.90 Obsolete status + obsolescence audit, §11.4.91 summary-doc clarity, §11.4.92 multi-pass change-evaluation, §11.4.93 SQLite workable-items SSoT, §11.4.94 zero-idle parallel-by-default) into the same agent-rule covenant cluster as short-form restatements citing the literal anchors 11.4.89/11.4.90/11.4.91/11.4.92/11.4.93/11.4.94, inherited per §11.4.35; restated + cited, not redefined; r9 propagates the next three inherited HelixConstitution mandates (§11.4.95 workable-items SQLite DB tracked-in-git-not-gitignored, §11.4.96 safe-parallel-work-with-long-build catalogue + mandate, §11.4.97 maximum-use-of-idle-time + progress-update cadence) into the same agent-rule covenant cluster as short-form restatements citing the literal anchors 11.4.95/11.4.96/11.4.97, inherited per §11.4.35; restated + cited, not redefined.
Continuation	bump again when the remaining open evidence-gated HRDs in docs/Issues.md close and the next release tag (post-v0.6.0) lands.

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Base agent rules: the Helix Constitution's AGENTS.md, provided by the **parent project's** constitution/ submodule (Herald does not carry its own copy).
READ IT FIRST.

Discover the constitution by walking up parent directories until you find `<ancestor>/constitution/Constitution.md`, or by invoking the canonical helper `<discovered>/find_constitution.sh`. For standalone Herald work, clone the constitution alongside Herald:

```
git clone git@github.com:HelixDevelopment/  
    HelixConstitution.git \  
    $(dirname "$PWD")/constitution
```

The base file is authoritative for any topic not covered here. Herald-specific rules below extend them; they never weaken them.

Canonical: <https://github.com/HelixDevelopment/HelixConstitution>

Critical base rules restated (for agents that don't follow @imports)

- **No bluffing.** Every PASS carries positive evidence. Constitution §11.4 / §1.1.
- **Mutation-paired gates.** Every new gate has a paired mutation proving it catches regressions. Constitution §1.1.
- **No guessing language** (likely, probably, maybe, seems, appears) when reporting causes. Constitution §11.4.6.
- **Credentials never tracked.** .env git-ignored; runtime-load only. Constitution §11.4.10. **Operator step-by-step guide for every supported messenger + dispatcher:** docs/guides/OPERATOR_CREDENTIALS.md (covers Postgres / Redis / Telegram HRD-011 / Claude Code HRD-012 + reserved env-var names for planned channels + audit checklist).
- **Never force-push.** Requires explicit per-session authorization.
- **Hardlinked backup before any destructive op.** Constitution §9.
- **60% RAM cap on heavy work.** Constitution §12.6.
- **Multi-upstream push.** Every commit fans out to all 4 hosts (GitHub + GitLab + GitFlic + GitVerse). Constitution §2.1.

Herald-specific agent rules

Project status (load-bearing for every task)

Herald is in **active multi-wave implementation** as of 2026-05-31 — waves 2 through 7 have shipped and the latest tag is **v0.6.0** (2026-05-28). The repo contains:

- README.md — mission, deployment model, constitution-inheritance contract, how to run.
- docs/specs/mvp/specification.V4.md — MVP spec (participant-attribution §11.4.104 + intent-recognition §11.4.105 incorporated; prior V-series archived under docs/specs/mvp/archive/).
- docs/guides/HERALD_CONSTITUTION.md — project-specific constitutional extensions.
- docs/guides/CONSTITUTION_INHERITANCE.md — operator/agent guide for the parent-discovery contract + the gate.
- upstreams/ — Herald's own mirror declarations (lowercase, §11.4.29-compliant).
- tests/test_constitution_inheritance.sh — inheritance gate (read-only assertions).

- `tests/test_constitution_inheritance_meta.sh` — paired mutation meta-test (§1.1).
- `.gitignore` tuned for Go (also ignores `.DS_Store`).

Herald **does not** ship a `constitution/` submodule of its own — see `docs/guides/CONSTITUTION_INHERITANCE.md` for the rationale and the discovery mechanism.

As of 2026-05-31 the codebase spans **18** Go workspace modules: 10 shared/foundation (`commons`, `commons_auth`, `commons_constitution`, `commons_infra`, `commons_messaging`, `commons_prefix`, `commons_storage`, `commons_tls`, `commons_watch`, `commons_workable`) + 8 flavor binaries (`pherald`, `sherald`, `cherald`, `bherald`, `rherald`, `iherald`, `scherald`, `qaherald`). The full §11.0 type contract is realized in `commons/types.go`. Live and tested: the `null://` sandbox adapter, the Telegram adapter (`getUpdates` long-poll + `send` + `reply` + `@-mention` render driving `pherald listen`), the Slack adapter (Wave 7), the Claude Code dispatcher (live `claude` invocation, Opus pinned, per-message timeout), `commons_storage` (`pgx` + `River` + `Redis`), the `Gin/v1/*` REST API, participant-attribution (§11.4.104), natural-language intent-recognition (§11.4.105), the Docs Chain documentation-sync mandate (§11.4.106), and the `commons_workable/commons_watch` ATMOSphere substrate (`pherald watch daemon`). On-demand container orchestration comes from the `containers/` submodule (`digital.vasic.containers`).

Build + test from repo root:

```
go build ./commons/... ./commons_prefix/... ./commons_messaging/... ./commons_
go test -race -count=1 ./commons/... ./commons_prefix/... ./commons_messaging/...
```

Still open (roughly a dozen evidence-gated HRDs in `docs/Issues.md` — live-integration credentials + later-iteration channels); the bulk of the catalogue migrated to `docs/Fixed.md` in the v1.0.0 closure sweep. Consult `docs/Issues.md` for the current open set before claiming a feature is unimplemented — HRD-010/011/012/016 (storage, Telegram, Claude Code dispatch, REST API) are all now LIVE.

When asked to "add a feature": find the spec section, open / claim the relevant HRD-NNN in `docs/Issues.md`, write Go + tests, ensure `go test` passes, close the HRD-NNN by migrating its row to `docs/Fixed.md` (per Universal §11.4.19 atomic-migration mandate).

Never invent build / test commands beyond `go test ./<module>/...`. Live-integration tests require operator-supplied bot tokens / Claude sessions / Postgres — `docs/CONTINUATION.md` carries the handoff prompt.

End-user-usability covenant (Herald §107 / Helix §11.4 — MANDATORY ANTI-BLUFF)

Forensic anchor — verbatim operator mandate:

"all existing tests and Challenges do work in anti-bluff manner - they **MUST** confirm that all tested codebase really works as expected! We had been in position that all tests do execute with success and all Challenges as well, but in reality the most of the features does not work and can't be used! This **MUST NOT** be the case and execution of tests and Challenges **MUST** guarantee the quality, the completion and full usability by end users of the product! This **MUST BE** part of Constitution of our project, its `CLAUDE.MD` and `AGENTS.MD` if it is not there already, and to be

applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well (if not there already)!"

Agent-binding rule. No agent — Claude Code, Codex, Cursor, Gemini, Aider, any CLI subagent — may declare a Herald feature "done", "landed", "shipped", or "tests pass" without **positive runtime evidence captured during execution** that an end user of the relevant <flavor>herald binary can actually use the feature.

- Unit-test green is necessary but **not sufficient**.
- Integration-test green is necessary but **not sufficient**.
- Compile success is necessary but **not sufficient**.
- A PASS line in a Challenge runner is a §11.4 PASS-bluff if it doesn't cross-check against runtime evidence (HTTP response body, DB row, log line, captured screenshot/audio, etc.).

Canonical Herald enforcement: `scripts/e2e_bluff_hunt.sh` — boots real services (Gin server + Postgres container via the `containers/` submodule), hits real `/v1/*` endpoints, asserts response bodies, runs M2 RLS-tenant-isolation integration tests, and SIGTERM-graceful-shutdowns. ALL 14 invariants MUST PASS before any release-tag, risky commit, or "milestone landed" claim. Canonical Helix authority: <discovered>/Constitution.md §11.4 + §11.4.1..§11.4.16. Canonical Herald authority: `docs/guides/HERALD_CONSTITUTION.md` §107. Inheritance-gate invariant **I8b** asserts the verbatim covenant anchor is present in this file.

Tests AND Challenges are bound equally: a Challenge that scores PASS on a broken-for-end-user feature is the same defect class as a unit test that does. Both are release blockers.

§107.x — docs/qa/ Evidence Mandate (operator mandate, 2026-05-22; cascades from Helix §11.4.83)

Forensic anchor — verbatim operator mandate (2026-05-22):

"every feature that ships MUST carry a recorded e2e communication transcript + any attached materials under `docs/qa/<run-id>/` (per-feature subdirectories). A feature with no QA transcript is itself a §107 PASS-bluff — it claims to work but has no auditable runtime evidence. Bot-driven automation (e.g. Herald's planned `qaherald` binary) MUST preserve full bidirectional communication threads as proof."

Agent-binding rule for Herald. No agent may declare a Herald feature done without `docs/qa/<run-id>/` evidence committed in the same logical work effort. Telegram-driven features (HRD-011, planned `qaherald`) capture both halves of the bot conversation; Gin `/v1/*` routes capture request + full response body; container-driven features (Postgres, Redis, OTel) capture container logs + healthcheck output. Bot-driven QA (the planned `qaherald` binary) MUST preserve the full conversation thread under `docs/qa/qaherald-<TS>/` — a stored-final-PASS-only mode is itself a §107 bluff.

CI / release gates MUST refuse to tag a release whose feature-shipping commits lack their matching `docs/qa/<run-id>/`. Canonical Helix authority: <discovered>/Constitution.md §11.4.83. Canonical Herald authority: `docs/guides/HERALD_CONSTITUTION.md` §107.x.

§107.y — Working-Tree Quiescence Rule (operator mandate, 2026-05-22; cascades from Helix §11.4.84)

Short tag: working-tree quiescence.

Forensic anchor — verbatim operator mandate (2026-05-22):

"no subagent commit may proceed while any concurrent mutation gate is in flight in the same checkout. Before `git add`, the committing agent **MUST** `grep` its own working tree for mutation markers (`MUTATED for paired, // always pass, return json.Marshal shortcut paths, etc.`). Any unexplained file in the staging area triggers **ABORT**."

Lesson (forensic — Herald-internal). Commit 72e81ab (logo fix subagent, 2026-05-21) swept a `// always pass` JWT-bypass mutation residue — left mid-cycle by a paired §1.1 Wave 4b mutation gate — into the unrelated commit, which was then pushed to all four mirrors. The SECURITY FIX d5bd360 ("restore commons_auth/middleware.go JWT verify (mutation residue in 72e81ab)") landed within the hour, but the production-equivalent-binary-with-bypassed-JWT window is a real security-defect window. The rule below is the constitutional outcome.

Agent-binding rule for Herald. Every commit flow (main thread + every dispatched subagent) **MUST**: (1) `grep` the working tree for mutation markers (`MUTATED for paired, // always pass, return json.Marshal shortcut paths, // MUTATION / # MUTATION annotations, _mutated_ * filename suffixes, .git/MUTATION_IN_PROGRESS lockfile`) **BEFORE** `git add`; (2) cross-check `git status --porcelain` against the declared scope — unaccounted files **ABORT**; (3) refuse to operate while `.git/MUTATION_IN_PROGRESS` is present (any active mutation gate **MUST** complete `mutate` → `assert FAIL` → `restore` → `assert PASS` **BEFORE** unrelated commits proceed); (4) prefer `git worktree add` per subagent when running concurrent subagent work.

Prototype. `tests/test_wave4b_mutation_meta.sh` carries the canonical Herald implementation (`check_quiescence()` at line 92; assertion at line 197). Generalising to all paired-§1.1 gates is open work. The planned universal scanner is `scripts/mutation_residue_audit.sh`.

Canonical Helix authority: <discovered>/Constitution.md §11.4.84. Canonical Herald authority: `docs/guides/HERALD_CONSTITUTION.md` §107.y. A mutation marker that lands in a tagged Herald commit is a critical defect regardless of how briefly it persisted — see 72e81ab / d5bd360 as proof.

Inherited covenant restatements — Helix §11.4.85 / §11.4.87 / §11.4.88 / §11.4.89 / §11.4.90 / §11.4.91 / §11.4.92 / §11.4.93 / §11.4.94 / §11.4.95 / §11.4.96 / §11.4.97 / §11.4.98 / §11.4.99 / §11.4.100 / §11.4.106

These twelve mandates are **inherited** from the HelixConstitution via parent-discovery (§11.4.35). Every agent — Claude Code, Codex, Cursor, Gemini, Aider, any CLI subagent — is bound by them; this file **restates** + **cites**, it does NOT redefine or weaken. The literal anchors are required by the §11.4.87 CM-COVENANT-114-87-PROPAGATION pre-build gate (asserts 11.4.85 / 11.4.87 / 11.4.88 present in every CLAUDE.md / AGENTS.md / QWEN.md).

§11.4.85 — Stress + Chaos Test Mandate (Helix, 2026-05-24). Agent-binding rule: no agent may declare a Herald fix or improvement done without full-automation **stress** (sustained / concurrent load) **AND chaos** (process-death / network-fault / input-corruption / resource-

exhaustion / state-corruption injection) test suites, each PASS citing a captured-evidence artefact under docs/qa/<run-id>/stress_chaos/ per §11.4.5 + §11.4.69. A happy-path-only PASS is a §11.4 / §107 PASS-bluff at the resilience layer. Binds Herald's flavor binaries (pherald listen under concurrent updates, Gin /v1/* under load, claude_code dispatch under process-death, container flows under disk/OOM pressure). Canonical authority: HelixConstitution Constitution.md §11.4.85 (inherited per §11.4.35).

§11.4.87 — Endless-loop autonomous work + zero-idle agent dispatch + anti-bluff testing (Helix, 2026-05-26). Agent-binding rule: when instructed to "continue in endless loop fully autonomously" (or equivalent), the agent MUST continue until ALL are simultaneously TRUE — Herald's loop checks docs/Issues.md Status-column (zero In progress/Ready for testing/In testing/Reopened per §11.4.15), docs/CONTINUATION.md §3 "Active work" empty, TaskList reports no subagent mid-execution, and no in-flight push/build/sync. The agent MUST dispatch background subagents for parallelisable non-contending work rather than serialise; idle is permitted ONLY while waiting on a result. Every closed item lands four-layer coverage (§11.4.4(b)) with real captured-evidence PASS; tests AND Challenges are bound equally. The loop terminates only on all-clear, explicit operator STOP, a §12 host-session-safety demand, or a scheduled wake against a known-future-actionable signal. No --idle-OK / --skip-endless-loop / --metadata-only-test-suffices escape exists. Canonical authority: HelixConstitution Constitution.md §11.4.87 (inherited per §11.4.35).

§11.4.88 — Background-push mandate (Helix, 2026-05-26). Agent-binding rule: every Herald commit flow MUST release the commit-lock (.git/.commit_all.lock) the instant git commit returns 0 — BEFORE any push — then spawn the push detached (nohup ./push_all.sh ... &+disown) with a per-remote flock (.git/.push.<remote>.lock) so same-remote pushes serialise while GitHub / GitLab / GitFlic / GitVerse push in parallel. The orchestrator's exit code reports COMMIT success, not push success. Backgrounded push failures land in qa-results/push_failures/<TS>_<remote>.log and the next autonomous-loop tick MUST surface them (silent push-failure is a §11.4 distribution-layer PASS-bluff). The ONLY synchronous-push escape is the explicit --sync-push flag for §11.4.41 force-push merge-first paths. Canonical authority: HelixConstitution Constitution.md §11.4.88 (inherited per §11.4.35).

§11.4.89 — Background test execution mandate (Helix, 2026-05-27). Agent-binding rule: any Herald test cycle the agent expects to exceed ~30s — the tests/test_wave*_mutation_meta.sh gates, scripts/e2e_bluff_hunt.sh, future stress/chaos suites — MUST be launched detached (nohup ... > qa-results/<test_id>_<TS>.log 2>&1 &+disown); the agent returns immediately to the §11.4.42 priority queue and polls the log / exit-status rather than blocking. Composes with §107.y / §11.4.84: a backgrounded mutation gate still mutates the shared tree, so the agent runs ONLY ONE against the main checkout at a time (serialised by the .git/MUTATION_IN_PROGRESS lockfile + scripts/mutation_residue_audit.sh pre-push scanner); concurrent gates require separate git worktree checkouts. Foreground is permitted only for <30s tests or on explicit operator request. Canonical authority: HelixConstitution Constitution.md §11.4.89 (inherited per §11.4.35).

§11.4.90 — Obsolete status + per-item obsolescence audit (Helix, 2026-05-27). Agent-binding rule: Herald's HRD Status closed-set gains a 4th terminal value Obsolete (→ Fixed.md) for items no longer valid (Reason closed-set: superseded-by-design-change / superseded-by-later-mandate / feature-removed / duplicate-of / unsupported-topology). The agent MUST add an **Obsolete-Details:** line within 8 non-blank lines of every Obsolete HRD heading: Since (ISO date), Reason, Superseding-item (§/HRD ref), Triple-check evidence (git-log / grep / runtime path per §11.4.6 — bare assertion forbidden). At every release-gate sweep the agent re-

evaluates every open + Fixed HRD for obsolescence; migrations are atomic per §11.4.19.
Canonical authority: HelixConstitution Constitution.md §11.4.90 (inherited per §11.4.35).

§11.4.91 — Summary-doc clarity (Helix, 2026-05-27). Agent-binding rule: every one-liner the agent writes in docs/Issues_Summary.md / docs/Fixed_Summary.md / docs/Status_Summary.md / README doc-link rows MUST be a self-contained clause (≥ 6 words OR ≥ 40 chars) naming the SUBJECT + the PROBLEM/GOAL — never a section-label fragment (Composes with, Closure criteria), bare metadata (Critical, Bug), a status restatement, or a bare HRD-id. Each is derived from the source long-form H1/H2 heading, never invented. Canonical authority: HelixConstitution Constitution.md §11.4.91 (inherited per §11.4.35).

§11.4.92 — Multi-pass change-evaluation discipline (Helix, 2026-05-27). Agent-binding rule: every non-trivial Herald change MUST pass — and the agent MUST document — a 5-pass evaluation before commit-ready: Pass 1 main-task captured-evidence (§11.4.5, no "should work"); Pass 2 regression-blast-radius (every touched file + every importer/caller audited); Pass 3 cross-feature interaction (shared state / timing / env — e.g. a gate edit checks §107.y quiescence + §11.4.89 backgrounding); Pass 4 deep-research validation (§11.4.8 — external precedent or literal "NO external solution found"); Pass 5 anti-bluff confirmation (no metadata-only / config-only / script-bug PASS). Evidence lands in the commit footer or docs/qa/ / qa-results/. Trivial changes (typo, revision-bump, MD-export regen touching zero source) are exempt only with explicit commit-message citation. Canonical authority: HelixConstitution Constitution.md §11.4.92 (inherited per §11.4.35).

§11.4.93 — SQLite-backed single-source-of-truth for workable items (Helix, 2026-05-27). Agent-binding rule: Herald's text-based HRD trackers (docs/Issues.md / Fixed.md / *_Summary.md / Status.md / CONTINUATION.md §3) migrate to a SQLite single-source-of-truth (docs/.workable_items.db, **version-controlled as Herald's authoritative SSoT** per operator mandate 2026-05-27 — a deliberate Herald divergence from the parent §11.4.93 gitignored-with-regeneration default, since for Herald the DB IS the authoritative artefact and MUST be committed, not regenerated-on-clone; only the transient WAL/SHM sidecars stay ignored) with bidirectional MD↔DB regeneration so sync-drift is mechanically impossible. The Go binary lives in the constitution submodule (constitution/scripts/workable-items/) — the agent references it per §11.4.74 catalogue-first, never reimplements. Migration is 6-phase; the agent files the tracking HRD (Phase 1) and progresses incrementally. Canonical authority: HelixConstitution Constitution.md §11.4.93 (inherited per §11.4.35).

§11.4.94 — Zero-idle priority-first parallel-by-default operating mode (Helix, 2026-05-27). Agent-binding rule (binding always-on contract): Herald work is NEVER idle while a priority-queued item can progress. Before any wake / sleep / "waiting for X", the agent surveys the priority queue, identifies all non-contending items, and dispatches them in parallel (subagent-driven per §11.4.20 / §11.4.70 when non-trivial; background per §11.4.89 when $> 30s$), picking highest-Severity/priority first. The conductor remains the integration + commit + push seam; parallel work MUST NOT compromise stability (composes with §107.y quiescence + §11.4.92 multi-pass + §12 host-safety). Idle is permitted ONLY when every item is genuinely blocked on external dependency, the operator issued STOP, or §12 host-session-safety demands it. Canonical authority: HelixConstitution Constitution.md §11.4.94 (inherited per §11.4.35).

§11.4.95 — Workable-items SQLite DB is TRACKED in git, never gitignored (Helix, 2026-05-27). Agent-binding rule: Herald ALREADY complies (operator correction 2026-05-27, recorded in §108.h + .gitignore) — the agent treats the workable-items SQLite DB as version-controlled, commits + pushes it alongside every state change, WAL-checkpoints it (PRAGMA wal_checkpoint(TRUNCATE)) before commit so only the transient -wal/-shm sidecars stay gitignored, and never force-rewrites it without a §9.2 hardlinked-backup. This is an explicit §11.4.30 carve-out and AMENDS the earlier §11.4.93 / §108.h "gitignored-with-

regeneration" text. The agent notes the alignment item: the constitution's canonical path is `docs/workable_items.db` (no leading dot); Herald's HRD-131 currently references `docs/.workable_items.db` — reconcile to the canonical path when the DB is implemented (HRD-131 Phase 2+, currently deferred). Canonical authority: HelixConstitution Constitution.md §11.4.95 (inherited per §11.4.35).

§11.4.96 — Safe-parallel-work-with-long-build catalogue + mandate (Helix, 2026-05-27).

Agent-binding rule: Herald has no AOSP build, but the principle binds — during ANY long-running Herald operation (a backgrounded mutation gate, `scripts/e2e_bluff_hunt.sh`, a §11.4.85 stress/chaos suite, a doc export) the agent MUST consult the safe-parallel catalogue and dispatch non-contending work rather than idle. SAFE-in-parallel for Herald: (A) MD/docs work, (B) `scripts/` helpers, (C) gate authoring, (D) test authoring, (E/F) commit + push to mirrors, (H) read-only analysis subagents, (I) web research, (J) workable-items DB ops. UNSAFE-during-a-running-gate (maps to §107.y): `git checkout / reset --hard / clean` on files a gate is transiently mutating, a SECOND concurrent mutation gate against the same checkout, host-session-safety breaches (§12). Subagent-driven default per §11.4.20 / §11.4.70. Canonical authority: HelixConstitution Constitution.md §11.4.96 (inherited per §11.4.35).

§11.4.97 — Maximum-use-of-idle-time mandate + progress-update cadence (Helix, 2026-05-27).

Agent-binding rule: (A) every minute of conductor idle time during which progressable, non-externally-blocked work exists is a violation — the agent dispatches work continuously through the whole idle window, not just at scheduled wakes. (B) The agent emits concise (1–3 line) operator-facing progress updates at milestone boundaries with NO prompt required: every HEAD advance (what landed), every subagent return (integrated), every constitutional anchor propagated, every captured-evidence artefact (`docs/qa/ / qa-results/` path), every Issues→Fixed/Obsolete closure. (C) Continuous physical-proof gathering per §11.4.5 / §11.4.6 / §11.4.69 — every closed item carries positive captured evidence committed alongside the closure. (E) Idle ONLY when genuinely blocked (operator STOP, external dependency, §12 host-safety). Canonical authority: HelixConstitution Constitution.md §11.4.97 (inherited per §11.4.35).

§11.4.98 — Full-Automation Anti-Bluff Mandate — Live tests MUST be re-runnable end-to-end without manual intervention (Helix, 2026-05-28).

Agent-binding rule: every test the agent writes, reviews, or marks PASSED under this Constitution — unit/integration/e2e/Challenge/stress/chaos/live — MUST be fully self-driving end-to-end with NO human action during execution (operator typing a Telegram message, hand-triggering a webhook, clicking a UI, attaching a file, anything beyond test startup → PASS/FAIL report). A test requiring manual action during execution is **by definition a §11.4 PASS-bluff at the automation layer**, regardless of how thorough the manual run is — it cannot run in CI, cannot validate regressions between manual runs, and the human dependency masks drift. Single permissible exception: one-time credential bootstrap OUTSIDE test execution (`.env` populated from a vault, shell exports in `~/ .bashrc`, OAuth approval at first install, MTPROTO session activation at first run — configuration, not test driving). Concrete Herald requirements: (1) no "operator MUST type a message" prompts in `tests/test_*.sh` or `_integration_test.go` — the agent drives programmatically (MTPROTO for Telegram, real-user-API for Slack, IMAP-test-account for email, webhook fixture, in-process loopback; never human keystrokes); (2) no `claude --resume <UUID>` against the same session UUID the dev conductor is using (Herald 2026-05-28 lesson: silent exit -1 collision — use a dedicated test-only UUID); (3) no 60s human-response windows (§11.4.50 determinism violation); (4) the agent verifies PASS at `-count=3` consecutive automated runs with self-cleaning state; (5) the agent classifies every existing test COMPLIANT vs NON-COMPLIANT in the §11.4.98 audit (release-gate item); (6) no false-positive PASS — silent-skip-as-PASS forbidden, stale-evidence forbidden, §11.4.3 SKIP-with-reason is correct. Currently NON-COMPLIANT Herald tests scheduled for MTPROTO-driven rewrite: `TestSubscribe_LiveBotAPI`, `tests/test_wave6_live_loop.sh`, Wave 6.5

lifecycle scenarios. Canonical authority: HelixConstitution Constitution.md §11.4.98 (inherited per §11.4.35).

§11.4.99 — Latest-Source Documentation Cross-Reference Mandate — instructions/guides/manuals MUST be verified against latest official online sources BEFORE publication (Helix, 2026-05-28). Agent-binding rule: every operator-facing instruction the agent authors or modifies — docs/requirements/blockers/*.md, docs/guides/*.md, README operator-action sections, troubleshooting cookbooks, OPERATOR_CREDENTIALS.md, MESSENGER_CHANNELS.md, TELEGRAM.md, HERALD_CONSTITUTION.md, integration setup walkthroughs — MUST be cross-referenced against the LATEST official online docs of every external service / library being documented BEFORE the commit lands. Misguidance-by-stale-docs is a §11.4 PASS-bluff at the documentation layer. Anchoring case study (Herald 2026-05-28): first-draft MTProto setup guide recommended VoIP + omitted the recover@telegram.org pre-login email; both contradicted Telegram's official docs + gotd/td maintainer guidance and could have permanently banned the operator's account. Concrete agent workflow: (1) fetch latest official docs via WebFetch / MCP / direct browsing — NEVER training data, NEVER memory, NEVER prior committed docs as the source of truth; (2) cross-reference each instruction against the source; (3) seek secondary authoritative sources (library maintainer SUPPORT.md, official changelogs, vetted community FAQs) when official source is sparse / silent; (4) cite source URLs + date in a `## Sources verified` document footer; (5) cite the cross-reference in the commit-message footer (`Sources verified <date>: <urls>`). (B) Negative findings documented (gaps / silences / contradictions never silently assumed authoritative). (C) Re-verification cadence: 6-month default staleness, 90-day for risk-classified services (messengers / cloud / payment / AI-LLM / code-hosting / package-managers); re-verify before operator-authority citation, at every vN.0.0 release, on breaking-change announcements, on operator-error report. (D) Risk-classified docs include explicit safety warnings vs latest policies. (E) Composes with §11.4.92 Pass 4 INDEPENDENT — cannot substitute. (F) Inheritance per §11.4.35, literal anchor 11.4.99. (G) Enforcement: missing `Sources verified` footer (doc + commit) → release-gate BLOCK; stale-beyond-grace → §11.4.90 Obsolete with Reason=stale-documentation. Canonical authority: HelixConstitution Constitution.md §11.4.99 (inherited per §11.4.35).

§11.4.100 — Video color + visual-quality fidelity mandate — every video-playback PASS MUST carry captured-frame deep-analysis (Helix, 2026-05-28). Agent-binding rule: when an agent authors or marks PASSED a video-playback test under this Constitution, the PASS MUST carry a captured-frame deep-analysis artefact proving rendered output matches host-extracted ground-truth source within tolerance — ΔE_{2000} + RGB/HSV histogram correlation + no pale/over-saturation/hue-shift + gamma/luma fidelity (BT.601/709/2020 + full-range-vs-limited-range explicitly), sharpness (Laplacian-variance), aspect-ratio, FPS+speed, continuity (no freeze, no drop-burst, no glitch, no obstruction-overlay per the §11.4.5 OCR census). Metadata-only PASS ("file exists" / frames>0 / codec registered) is forbidden; comparison harness MUST itself be golden-pair + bad-pair self-validated (no false-positive AND no false-negative). **Herald applicability:** non-applicable-but-cite — Herald has NO video-playback surface (pherald listens/downloads video attachments as opaque sha256-content-addressed blobs via the HRD-005 attachment pipeline and never decodes / renders / displays); Claude Code dispatcher receives the file path as text reference, not a rendered frame. The mandate binds latently — if Herald ever ships a video-rendering surface (preview thumbnails, in-app player, inline diff viewer), the agent applies §11.4.100 in full at that point. Cascade-parallel: §11.4.96 ("Herald has no AOSP build, but the principle binds") — same restatement-with-non-applicability-but-cite shape. Inheritance per §11.4.35, literal anchor 11.4.100, required by upcoming CM-COVENANT-114-100-PROPAGATION pre-build gate. Canonical authority: HelixConstitution Constitution.md §11.4.100 (inherited per §11.4.35).

§11.4.106 — Docs Chain documentation-sync mandate — derived doc artefacts MUST be kept in sync via the shared Docs Chain engine, never ad-hoc (Helix, 2026-05-29). Agent-binding rule: the documentation-sync obligations (Issues_Summary §11.4.12, Fixed_Summary §11.4.53, Status/Status_Summary §11.4.45/§11.4.56, README doc-link + export §11.4.57/§11.4.59, universal md→html/pdf export §11.4.65, roster fingerprint §11.4.86, workable-items DB↔MD §11.4.93/§11.4.95, CONTINUATION exports §12.10) are mechanized by **Docs Chain** — a content-hashed incremental DAG sync engine (`digital.vasic.docs_chain`) consumed BY REFERENCE via the constitution submodule (Phase-6 distribution operator-gated per §11.4.66). The agent registers chains in `.docs_chain/contexts/*.yaml` and runs `docs_chain sync/verify`; the engine **refuses to fake success** (tool-absent → rollback, exit 3) and **refuses silent merges** (both-dirty → conflict, exit 2) — §11.4.6 no-guessing + §11.4.50 determinism made mechanical, not aspirational. **Herald applicability: APPLICABLE (integration in progress)** — Herald's ~76 markdown docs + their html/pdf/docx siblings (today via the `regen-all scripts/export_docs.sh`) migrate to `docs_chain exec`: transforms that preserve Herald's exact logo-CSS pandoc flags (byte-compatible, zero regen-drift across ~223 siblings) per [docs/research/docs_chain/HERALD_DOCS_CHAIN_PLAN.md](#); planned `verify` drift-gate `e2e_bluff_hunt` invariant E146. Pending: Phase-4 CLI (landed 2026-05-31), the `exec`-staging relative-asset contract (gap G6), the binary-hash `verify` fix (dogfood finding — binary kinds hashed inconsistently between `sync-record` and `verify-check`). Cascade-parallel: §11.4.96 ("Herald has no AOSP build, but the principle binds"). Inheritance per §11.4.35, literal anchor 11.4.106, required by the CM-COVENANT-114-106-PROPAGATION pre-build gate. Canonical authority: `HelixConstitution Constitution.md` §11.4.106 (inherited per §11.4.35).

Participant identity, attribution & notification-tagging (Herald §109 / contract `docs/design/PARTICIPANT_ATTRIBUTION.md` — inherited per §11.4.35)

Agent-binding rule (operator mandate, 2026-05-31). Every messenger must relate each message to a **Participant** (a logical Subscriber/User — one person/agent, with a potentially DIFFERENT username on every messenger); workable items gain `created_by` + `assigned_to`; outbound notifications **@-tag** the right participant per a fixed rule matrix. The **single authoritative contract** every agent codes against is [docs/design/PARTICIPANT_ATTRIBUTION.md](#). These rules are restated (root definitions) in `HelixConstitution Constitution.md` / `CLAUDE.md` / `AGENTS.md` / `QWEN.md` and inherited here per §11.4.35 — the agent restates + cites, never redefines or weakens.

- **Identity model.** Backed by PG subscribers (logical party: canonical messenger-neutral handle, `display_name`, `kind` ∈ {human, agent, service}) + `subscriber_aliases` (per-channel `subscriber_id`, `channel`, `channel_user_id`, + **NEW** username TEXT — the per-channel @handle for tagging, distinct from `channel_user_id`). Canonical handle (stored in `created_by/assigned_to`) ∈ {Claude (reserved system-agent sentinel, `kind=agent`, NEVER tagged), a human's canonical handle (defaults to their Telegram @username; messenger-neutral; per-channel @usernames resolve via `subscriber_aliases`)}.
- **Operator env var.** The one human driving the system via the Claude Code CLI is designated by `HERALD_TGRAM_OPERATOR_USERNAME` (e.g. `@milos85vasic`), generalizing to `HERALD_<CHANNEL>_OPERATOR_USERNAME` — env var, NOT a DB flag. The operator's canonical handle = that env value; the operator is a normal Participant.
- **Attribution.** `created_by` = `OperatorHandle()` when opened via the Claude Code CLI prompt; "Claude" when System/Claude detects the issue/task/improvement/missing-feature; the sender's resolved canonical handle when received through Herald.

assigned_to defaults to OperatorHandle(), overridable explicitly. Both store the canonical handle string.

- **Tagging matrix.** On any item event, the notification to each channel @-mentions: assigned_to if human AND ≠ Operator; created_by if human AND ≠ Operator AND ≠ "Claude". "Claude" and the Operator are NEVER tagged (no system-ping, no self-ping). De-dup, then resolve each handle to that channel's @username (skip if no alias on that channel). The agent ships unit + integration + E2E + full-automation tests with real captured evidence under docs/qa/<run-id>/ — the matrix proven by a per-cell truth-table test + a one-cell-flip mutation that must FAIL, plus a NEGATIVE case proving the Operator is NOT tagged (§107 / Helix §11.4 anti-bluff). Canonical authority: HelixConstitution root §participant-attribution (inherited per §11.4.35); Herald contract docs/design/PARTICIPANT_ATTRIBUTION.md.

Intent recognition & clarification (Herald §110 / contract docs/design/INTENT_RECOGNITION.md — inherited from HelixConstitution §11.4.105 per §11.4.35)

Agent-binding rule (operator mandate, 2026-05-31). Subscribers must NOT need to know command syntax — there is no COMMAND: prefix and no fixed grammar. A subscriber sends a clear natural-language message and **the System determines the intent**. The **single authoritative contract** every agent codes against is [docs/design/INTENT_RECOGNITION.md](#). These rules are restated (root definitions) in HelixConstitution Constitution.md / CLAUDE.md / AGENTS.md / QWEN.md as §11.4.105 (the root-§ being added on the constitution stream) and inherited here per §11.4.35 — the agent restates + cites, never redefines or weakens.

- **Three-tier resolution (first tier that succeeds wins). Tier 1 — command recognition:** a deterministic CommandRecognizer (pherald/internal/inbound) maps a clear natural-language imperative to a structured action WITHOUT an LLM round-trip (fast-path, no prefix). It is CONSERVATIVE — only a confident match fast-paths; when in doubt it returns "no match" and defers to Tier 2. **Tier 2 — intent inference:** when no command matches, the Claude Code dispatch (the LLM) infers intent and returns a <<<HERALD-REPLY>>> action; the <<<HERALD-DISPATCH-v1>>> envelope INSTRUCTS the LLM to recognize Herald's command set, map natural language to the right action, and NEVER guess. **Tier 3 — clarify (fallback):** when neither a command nor a confident intent can be determined, action="clarify" — the System REPLIES to the original message, TAGS the sender (@username, via the §11.4.104 IdentityResolver), and asks a precise clarifying question naming the candidate intents.
- **Never guess, never ignore.** A wrong action is worse than a clarifying question (§11.4.6 no-guessing); the subscriber is never silently dropped and never has to learn syntax. The agent ships unit + integration + E2E + full-automation tests with real captured evidence under docs/qa/<run-id>/ — a Tier-1 truth-table including the conservative negatives that MUST fall through to "no match", a Tier-3 E2E asserting the reply body is EXACTLY @<sender> <specific question>, and a paired §1.1 mutation (break the confidence guard or drop the clarify tag) that must FAIL (§107 / Helix §11.4 anti-bluff). Canonical authority: HelixConstitution Constitution.md §11.4.105 (inherited per §11.4.35); Herald contract docs/design/INTENT_RECOGNITION.md.

Inheritance gate (run before any commit that touches root docs or constitution/)

```
bash tests/test_constitution_inheritance.sh          # the gate
bash tests/test_constitution_inheritance_meta.sh     # paired
              mutation proof (§1.1)
```

Both MUST return 0. If either fails, fix at root cause per Constitution §11.4.4 — never silently accept the FAIL.

Spec-change rule (load-bearing — docs/specs/mvp/specification.V4.md §"Specification documents")

Whenever docs/specs/mvp/specification.V4.md or any file under docs/specs/ (any depth) is modified, **comprehensive planning and implementation of all changes is MANDATORY** — agents may not edit the spec in isolation. This rule does NOT apply to creating or renaming files; for those, ask the operator what to do with the new path. Treat every spec edit as a project-wide ripple, not a doc tweak.

This rule is mirrored in CLAUDE.md and in docs/guides/HERALD_CONSTITUTION.md §106. The inheritance gate's invariant **I7a–c** asserts the rule anchor (comprehensive planning and implementation) is present in all three files; a missing copy is a §1.1 propagation bluff and the gate FAILs.

Multi-host mirror convention (Herald's own upstreams)

upstreams/ contains one script per mirror host (GitHub, GitLab, GitFlic, GitVerse). Each script exports UPSTREAMABLE_REPOSITORY and is sourced, not executed. The Herald repo's origin remote is already fan-out (1 fetch URL + 4 push URLs) — a single `git push origin <branch>` propagates to all four hosts. Per-host naming intentionally matches each provider's brand capitalization; do not normalize.

Forbidden in this project

- **Re-adding a constitution/ submodule inside Herald.** Herald is consumed as a submodule of a parent project that already provides constitution/. A duplicate copy inside Herald is a deployment-model violation. The inheritance gate's invariant I6 enforces this — re-adding it will turn the gate red.
- Promoting Herald-specific values into the parent constitution (universal status must be EARNED; see Constitution §11.4 + §11.4.10).
- Modifying any file under the discovered constitution/ from Herald commits — the parent's constitution is read-only from Herald's perspective. Constitution changes go through the HelixConstitution repo directly.
- Adding new submodules without re-running `bash tests/test_constitution_inheritance.sh` afterward.